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Hierarchical Attention-Based Astronaut Gesture Recognition: A Dataset and CNN Model

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ABSTRACT In the space station, visual gesture recognition is an important component for gesture based human-robot interaction to control the robot, which assists astronauts to complete simple, repetitive and cooperative work. However, public gesture datasets are captured in the daily environment and existing approaches trained on these datasets achieve imprecise performance for astronaut gesture recognition. In this paper, we introduce a new astronaut gesture dataset (DSSL-Astronaut gesture dataset) and a novel hierarchical attention single-shot detector network (HA-SSD) for astronaut gesture recognition. Specifically, this dataset consists of the real and augmented images. The real images captured in the simulated space station are closed to the real images in the space station environment. Meanwhile, we utilize the Mask-RCNN model to segment the foreground image including an astronaut from real data and capture the background images of the simulated space station at different views and illuminations. Then we combine them to synthesize the augmented images. A novel HA-SSD model consists of a lightweight backbone named MobileNet and a hierarchical attention mechanism. The MobileNet is used as the feature extractor, which contains several depth-wise separable convolutions to make a trade-off between latency and accuracy. Meanwhile, the hierarchical channel-wise attention module is adopted to exploit the fine semantic information to enrich the features for improving the performance of gesture recognition. We conduct experiments to demonstrate that our dataset is suitable for the space station and our approach is able to effectively localize and recognize the gestures with strong generality.

INDEX TERMS Assist robot, attention, gesture dataset, gesture recognition, single-shot detector.

I. INTRODUCTION

A. BACKGROUND

The space station is an important platform for carrying out experiments in the field of aerospace science and technology. In order to maintain the stable operation of the space station, astronauts need to do lots of work, including some simple, repetitive and cooperative work. With the rapid process of artificial intelligence and human-computer interaction technology, it is completely feasible to design an intelligent robot to assist the astronauts for accomplishing these tasks. Such robots need to have strong autonomy and intelligent interaction between robots and astronauts. To achieve this goal, we propose an assist robot named Intelligent Formation Personal Satellite (IFPS) [1], [2], for astronaut assistance in the space station cabin.

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A variety of robots have been proposed to assist astronauts to work in the space station, and many key technologies about these assist robots have been explored. NASA proposed the idea of a Personal Satellite Assistant (PSA) [3] in 1998. PSA was supposed to possess the ability to interact with astronauts by utilizing the information of voice and human movement, but the project lapsed and PSA has never been used on the space station. Synchronized Position Hold Engage and Reorient Experimental Satellite (SPHERES) [4] was successfully applied to the international space station in 2006, providing scientists with a zero-gravity, three-dimensional mobile test platform. Internal Ball Camera (Int-ball) [5] was developed by the Japan aerospace exploration agency and has been put into use on the international space station in 2017. Astrobees [6] was sent to the international space station in 2019, which supported remote control or manual control, and was used to read mission sensor data and monitor the activities of all crew members. The purposes of these robots are to improve the efficiency of astronauts in orbit.



FIGURE 1. Intelligent Formation Personal Satellite (IFPS).

Although the existing assist robots in the international space station have made some breakthroughs, they are not capable of complex human-robot interaction. As shown in Fig.1, it is a 2.3-kilogram sphere with a diameter of 260 millimeters. Based on an RGB-D camera and IMU, IFPS can complete visual navigation and avoid obstacles with 6 sonar sensors. Compared to the four assist robots mentioned above, a major contribution of IFPS is the intelligent ability to interact with astronauts. To be consistent with the real applicable scenario as much as possible, we build a ground simulation cabin that is similar to the environment of the real space station cabin as shown in Fig.2. Our team has proposed many researches in this cabin, such as astronaut visual tracking, visual based in-cabin navigation et al [1], [2]. In our scenario, IFPS independently has the ability to fly in the cabin and follow its own “master astronaut”. The astronaut would make the specific gestures of left or right hand which are captured by IFPS. Then, IFPS would detect and recognize these gestures by the captured images towards performing the corresponding tasks. There are several challenges in the astronaut gesture recognition task: accurately locating and identifying astronauts’ gestures from a complex background in the cabin of space station, the different hand shapes, the interferences of non-command hand and the lack of proper astronaut gesture dataset.

B. CONTRIBUTION

Firstly, a new dataset (DSSL-Astronaut gesture dataset) with 7 categories of the gesture are provided for astronaut gesture recognition, which consists of the real images and augmented images. To our best knowledge, this is the first gesture recognition dataset tailored to the context of space missions. We built a ground simulated space station, then engaged volunteers to make different gestures in it and captured images to get the real images. As to the augmented images, we utilize

the Mask-RCNN [7] model to segment the foreground image including an astronaut from real data and capture the background images of the simulated space station at different views and illuminations. Secondly, we propose a new model called HA-SSD which consists of a lightweight backbone: MobileNet-SSD [8] and a hierarchical attention module. HA-SSD out-performs original SSD [9] in Oxford hand dataset [10] and DSSL-Astronaut gesture dataset. In addition, our approach shows better generality than original SSD and YOLOv3 [11].

C. PAPER ORGANIZATION

This paper is organized as follows: In section II, we provide recent related works of gesture recognition approaches. Section III introduces DSSL-Astronaut gesture dataset. In section IV, we describe the architecture of the astronaut gesture recognition network. In section V, we introduce experiments and report the analyzed results. And in section VI, we summarize the findings and contributions of this study.

II. RELATED WORK

A. CONVENTIONAL GESTURE RECOGNITION APPROACHES

Many approaches have been explored in the field of gesture recognition. Some researches on gesture recognition attempt to use wearable devices such as data gloves [12] or electrode-based systems [13], [14]. However, this form of recognition is inconvenient and presents a limitation of human movement. Most gesture recognition studies based on vision [15]–[18] use the skin color feature or hand shape as the guidance of segmenting the hand area from the whole image, and then use them to conduct template matching for gesture recognition. These methods get rid of the shackles of devices and make human-computer interaction more natural, but they are all vulnerable to skin colour and ambient light. Hence, this method is not proper to the aerospace field which requires surefire performance of recognition.

B. OBJECT DETECTION METHODS BASED ON DEEP LEARNING AND LIGHTWEIGHT NETWORK

Deep learning improves the ability of approaches to find optimal features and handle indirect relationships between features and goals. It automates feature engineering to obtain more robust features by learning-based way in an end-to-end manner. In recent years, with the rise of artificial intelligence [19]–[21] and human-computer interaction technology [22]–[24], object detection and Gesture recognition methods based on deep learning have made innovative breakthroughs [25]–[27], e.g. RCNN [28], Fast R-CNN [29], SSD [9], YOLO [30], etc. Which have shown outstanding performance in the field of object detection. Lots of previous work improve these approaches for gesture recognition and get good applications [31]–[36]. This section briefly reviews existing works belonging to two major categories:

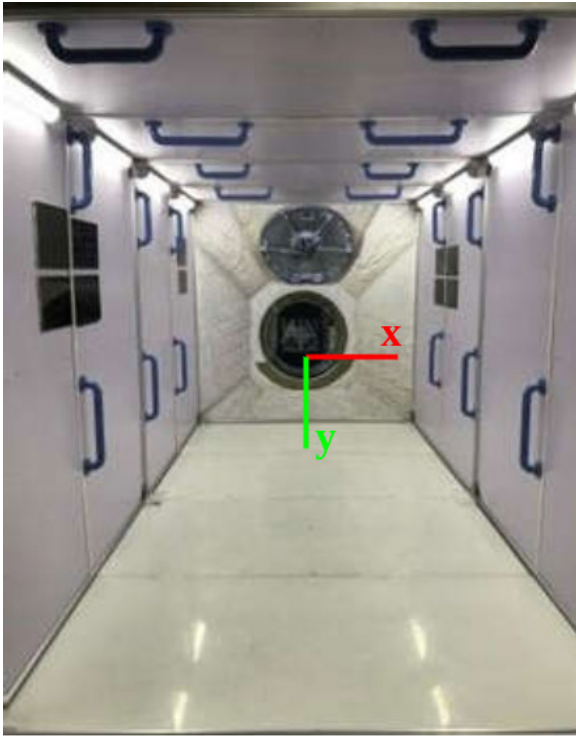


FIGURE 2. The front viewpoint in the simulation cabin of the space station. Astronauts gestures motions are restricted to x,y planes.

two-stage detector and one-stage detector. Two-stage detector has higher performance but slower speed than one-stage.

RCNN generates potential bounding boxes in an image according to handcraft features, and then train a classifier based on normalized proposed boxes, followed by post-processing to eliminate duplicate results. Fast R-CNN feeds an image into CNN to acquire feature maps, and a selective search method is adopted to extract region proposals of the image. Afterwards, it uses region of interest (ROI) pooling to generate every region proposal patch in feature map as the input of predicted layer. Fast R-CNN dramatically speeds up detecting process and improve the performance of detector. Single-shot detectors have shown to provide simultaneously excellent performance and real-time speed.

YOLO divides the input image into grids, in which each cell within grid predicts some bounding boxes and the corresponding confidence scores. It only predicts one set of class probabilities per grid cell regardless of the number of boxes. Hence, it is extremely fast with state-of-the-art performance in detection accuracy. SSD produces a fixed-size collection of bounding boxes and the score for the probability of an object class in the boxes. A non-maximum suppression is adopted to produce the final detections after classification. SSD adopts multi-scale feature maps default boxes for object detection in different size to improve detection accuracy.

Meanwhile, lightweight network framework such as MobileNet [8], [37], [38] were proposed to explore efficient model designs in the classification task. Hence,

the combination between MobileNet and SSD is more suitable for our gesture recognition task, which is able to make an effective tradeoff between latency and accuracy.

C. GESTURE DATASET AND DATA AUGMENTATION

Many datasets were established for gesture recognition [39]. ChaLearn LAP IsoGD Dataset contains 47933 RGB-D gesture videos [40], 2013 Chalearn Gesture Challenge dataset consists of 11000 images with 20 categories [41], and Oxford hand dataset with 13050 images, which collect hand image from various public image sources. EgoGesture [42] contains more than 24000 gesture samples and 3000000 frames for both color and depth in the first-person view. All datasets mentioned above are not proper for astronaut gesture recognition due to they are lack of space station environment.

Overfitting is a serious problem in deep learning, which leads to an excellent performance on the training process, but a poor performance in the test process. It can be effectively avoided through data augmentation. Existing approaches demonstrated the effectiveness of data augmentation through some techniques. A data augmentation approach named Random image cropping and patching (RICAP) [43] randomly crops four images and patches them to synthesize a new image, and mixes the class labels of the four images to get an advantage of the soft labels. Reference [44] divides the vibration data into samples by segmentation, then recombines samples to increase the number of images.

III. DSSL-ASTRONAUT GESTURE DATASET

A. PROPERTIES FOR ASTRONAUT HAND DATA

In the space station cabin, IFPS flies around its own astronaut master and captures the “control hand” at any time. After obtaining the specific gestures of “control hand” from the astronaut, IFPS executes the corresponding command to complete the whole process of human-computer interaction. Based on this applicable scenario, our astronaut gesture dataset should meet the following demands:

Simulated Cabin Scene: IFPS will be applicable in space station cabin in the future, so it is necessary to simulate the real cabin scene and even simulate the astronauts to collect hand gesture images.

Complex Background: IFPS will fly around the space station, so it will capture astronauts in different positions. It is necessary to simulate complex background under different angles, light and so on.

Object Diversity: Due to the impact of different colors of clothing on different astronauts, the dataset should be captured by images including multiple people and diverse clothing.

Gesture Class: The gestures that are used in the interaction between robots and humans should be meaningfully and easy to remember by the astronauts. In order to identify different commands more accurately, the differentiation of each gesture should be as large as possible. Suitable gesture definitions are convenient for common users which could

TABLE 1. DSSL-astronauts gesture and means.

Confirm	Down	Up	Left	Right	Stop	Background
						
After IFPS detect other commands, it should detect the "confirm" before it executes instruction.	If the "down" gesture is detected, after confirming instruction, IFPS should fly 10cm down.	If the "up" gesture is detected, after confirming instruction, IFPS should fly 10cm up.	If the "left" gesture is detected, after confirming instruction, IFPS should fly 10cm left.	If the "right" gesture is detected, after confirming instruction, IFPS should fly 10cm right.	If the "stop" gesture is detected, after confirming instruction, IFPS should stop its current mission and return to its charging station.	Meaningless gesture.

reduce the extra learning time for gestures. Under these principles, we defined 6 gesture commands with special meaning that conform to sign language custom. The gesture dataset includes 6 gesture commands: "fly up", "fly down", "fly left", "fly right", "confirm" and "stop".

IFPS always flies around and detects astronauts expect for charging at the charging station. So, it is possible for IFPS to capture unintentional gestures without any meaning. To alleviate the influence of the meaningless gestures, a gesture "background" without any instructions is defined to contain the redundant gestures. In space station, astronauts may inadvertently make gestures that IFPS captures and recognizes as meaningful gestures. But the probability of capturing the "confirm" gesture after misrecognition is almost zero.

As shown in table 1, we design 7 categories of gestures which cover most kinds of basic manipulation commands. Gestures: Down, Up, Left, Right simply enable low-level interactions between astronauts and IFPS. When performing repetitive and cooperative tasks, gestures: Confirm and Stop are extremely useful to clarify the beginning and end of tasks.

Distraive Data: There could be several astronauts in the space station cabin, and all of the astronauts' hands could be captured by IFPS, which would need to locate the "control hand" inside. Our previous research [1] proposed a Kalman filter based method to track the master astronaut. We need to further investigate the methods that can confirm the "control hand". Therefore, our data should include the image of two or more hands in the picture, i.e. distractive data.

B. DATA COLLECTION

Three volunteers are recruited to perform different gestures in the simulation space station cabin one by one. To collect gesture data, we select Camboard pico maxx camera to record video in the resolution of 640×480 with the frame rate of 30 fps. This camera is controlled within a range of 1.4 – 1.6 m from volunteers. After recording all videos, we extract 5 frames per second in the videos to get the static gesture images. We define 4 scenes:

(1) Different costumes: every volunteer is asked to perform gestures in blue, indigo and green costumes, respectively.

We choose these three colors because they are commonly used in penguin suits.

(2) Left hand and right hand: volunteers perform gestures by using their left hands first, then they are asked to make the same gestures by their hands.

(3) Standard gestures: except for standard gestures, volunteers are asked to make some vertical offset of the hand when performing the "up" gesture and the "down" gesture. in the meanwhile, they make some horizontal offset of the hand when performing the "left" gesture and the "right" gesture. Some hand tilts are performed in the vertical direction when they perform the "confirm" gesture and the "stop" gesture.

(4) Clear images and motion-blurred images: volunteers perform gestures at slow speed first, then they are asked to make the same gestures in high speed. In addition, volunteers are asked to perform meaningless gestures, all of these meaningless gestures were defined as "background" gesture.

C. DATA AUGMENTATION

Data augmentation can avoid overfitting which is a serious problem in deep learning. In astronaut gesture recognition, we explore and compare multiple solutions to the problem of data augmentation. However, in these methods, a random cropping method may cut out the hand in an image, while random flip may lead to command change such as the flipped "left" hand is similar to the "right" hand. Hence, we adopt a kind of general method based on image segmentation to synthesize the data.

With the development of image segmentation, existing methods are able to segment the complete objects in an image. Mask R-CNN [7] is a simple and general framework for object instance segmentation, which efficiently detects objects in an image while generating a segmentation mask simultaneously for each object. Especially, Instance segmentation is mature in person segmentation, so it is suitable to use this approach to extract the astronaut from a digital image.

Given a real image $x \in \mathbb{Z}^{w \times n \times c}$, we use Mask R-CNN as an off-the-shelf instance segmentation method to extract a mask

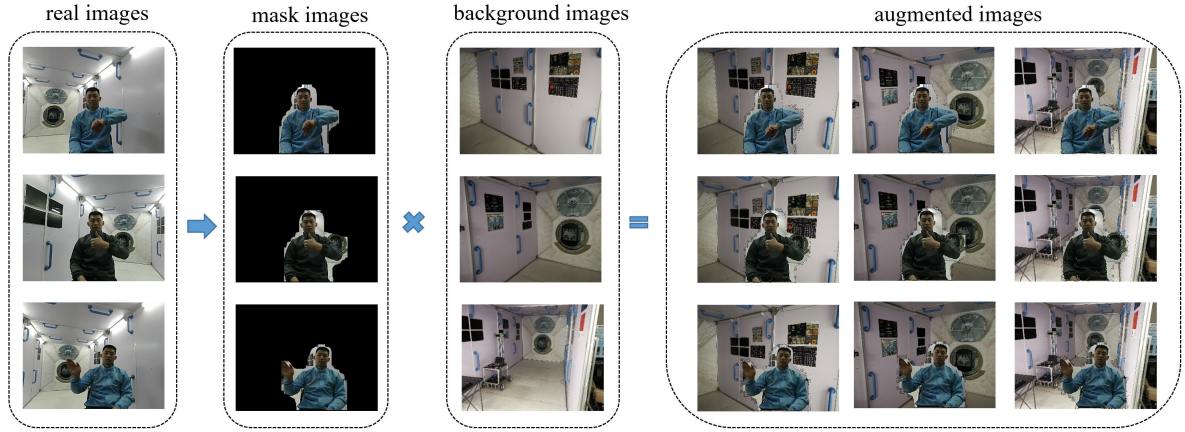


FIGURE 3. The process of data augmentation. We segmented the astronaut from real image as the foreground image and captured simulated space station cabin in different views and different illuminations as the background images. In the last, we synthesize new images by combining the foreground and the background images.

map:

$$m = f(x) \in \mathbf{Z}^{w \times n \times c}, \quad (1)$$

where f represents to the Mask R-CNN method and $m \in \{0, 1\}^{w \times n \times c}$ refers to the mask map for masking the astronaut in an image. Then, we make the logical AND operation between the original image and the mask map for generating the person map:

$$p_i = x \& m \in \mathbf{Z}^{w \times n \times c}, \quad (2)$$

where $\&$ refers to the logical AND operation. After consuming the person map, we capture some background images without the persons in the multi-view and multi-illumination cabins and obtain n background images $x_g = [x_{g1}, \dots, x_{gn}]$. After obtaining the different astronauts and backgrounds, we combine the x_g and p_i to generate the n synthetic images as Fig.3 shows. The above method efficiently augments this dataset under the premise of retaining the intact gestures. Meanwhile, it is able to be adaptive for variation in background and illumination. Although augmented images can be easily identified, they serve as a data augmentation method and eventually contribute to our gesture recognition algorithm.

D. DATASET STATISTICS

3 volunteers are invited to perform 7 classes of gestures in 4 different scenes. We collect 528 videos, and then extract totally 9126 real images from these videos. To get more data, we choose 300 real images for extracting the astronauts from them and capture 20 images of simulated space station cabin in different views and illuminations as the background image. As a result, we create 6000 synthetic images by combining them. 13858 images include all 6000 augmented images serves as training data, while the rest 1268 images serve as testing data. Table2 summarizes the character of the dataset, including the number of real images, the colour of costumes and different volunteers. We also introduce the tolerable max

TABLE 2. Summary of DSSL-astronaut gesture dataset.

Commands	7
Real image	9126
Composite images	6000
Costume colors	3
Persons	3
Hands	Right & left
Max offset (vertical)	[-0.2,0.2] width
Max offset (horizontal)	[-0.2,0.2] height
Max tilt (vertical)	[-60,60] degree

offset of the hand along with vertical or horizontal, and the max tilt in vertical. The maximum limitation of the offset and tilt are defined to distinguish invalid gesture. For example, a valid “right” gesture is a fist with the thumb pointing to the right in front of the body. If the distance between the center of the astronaut and his/her fist is more than 0.2 image width, it is identified as a kind of invalid gesture.

Even though many gesture datasets have been proposed for gesture recognition, our DSSL-Astronaut gesture dataset is still significant because of the simulate space station environment which is closed to the real environment.

IV. HIERARCHICAL ATTENTION SINGLE-SHOT DETECTOR

In this section, we elaborate on the proposed architecture named hierarchical attention single-shot detector as shown in Fig.4, in which this detector contains three components: the backbone based on MobileNet, the standard SSD scheme, and the hierarchical attention scheme. The lightweight MobileNet is designed by the point-wise and depth-wise convolutional operators for reducing the parameters of model and maintain the ability to extract the robust feature maps which are used

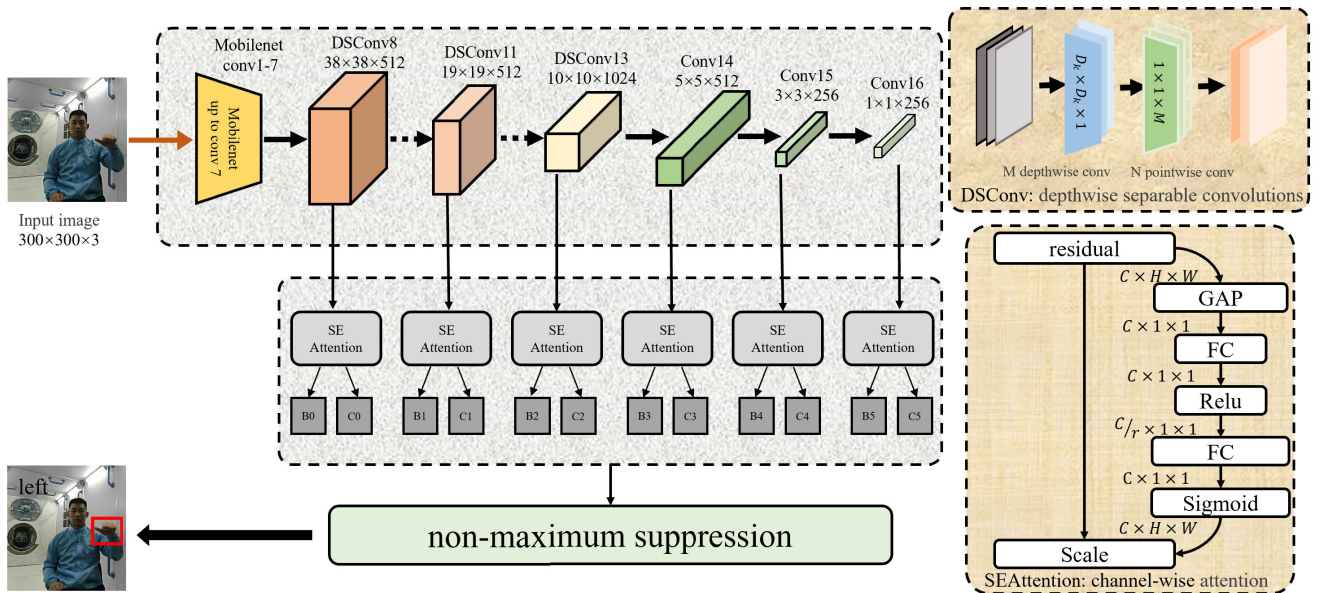


FIGURE 4. The framework of HA-SSD. It adopts the 1 to 13 convolutions of MobileNet to extract global features, 6 HA building blocks based on SE attention to learn hierarchical features. All the features are feed into non-maximum suppression module for getting the final result.

for the subsequent scheme. After obtaining the multi-scale contextual feature maps from the MobileNet, the hierarchical attention scheme refines these feature maps by channel-wise attention mechanism, towards focus on the salient regions. The standard SSD scheme discretizes the enhanced feature maps corresponding to the bounding boxes into a set of default boxes over different aspect ratios and scales.

A. CHALLENGES OF ASTRONAUTS GESTURE RECOGNITION

In our task, astronaut gesture recognition and tracking have the following difficulties:

Real-Time: IFPS is expected to interact with astronauts in real-time, so our recognition algorithm needs to consider the tradeoff between accuracy and speed.

The Problem of Detection: The different distances between the astronaut hands and IFPS may cause that the sizes of gesture objects in the images are different.

The Problem of Recognition: Different from general object recognition, astronaut gesture recognition should pay more attention to the fine-grained feature of hand. Whether a finger is detected or not may lead to different recognition results, so the detection bounding boxes must perfectly enclose the entire hand of the astronaut.

B. BACKBONE

We adopt an efficient model called MobileNet as the feature extractor, which contains a lightweight operator named depth-wise separable convolutions to efficiently make a tradeoff between latency and accuracy. The main component of MobileNet is depth-wise separable convolutions which factorize a standard convolution into two operators: a 3×3

depth-wise convolution and a 1×1 point-wise convolution. This factorization is able to drastically reduce computation and model size, meanwhile which retains the performance of the model. Inside, the novel depth-wise convolution layer composes of a single filter which convolutes the input feature maps to output single channel output while the point-wise convolution applies a 1×1 convolution layer to combine the outputs of the depth-wise convolution layer.

By exploring network topologies, we adopt an efficient network structure. The whole backbone structure is shown in table 3, which is based on the depth-wise separable convolutions. All layers in the table 3 are followed by a batchnorm (BN) layer [45] and ReLU nonlinearity function. Counting depth-wise and point-wise convolutions as separate layers, MobileNet has 26 layers. We utilize the outputs of DSCov8, DSCov11, DSCov13, Conv14, Conv15 and Conv16 layers for subsequent detection, in which feature sizes are 38×38 , 19×19 , 10×10 , 5×5 , 3×3 and 1×1 respectively.

C. HIERARCHICAL ATTENTION MODULE

Despite the standard SSD framework preserving the multi-scale semantic information, it still lacks the ability of locating the fine-grained hand gestures. So we adopt an attention mechanism to boost the representation of the salient hand region in the images. [46] demonstrate that the channels of feature highlight the representation of different region of image. Hence, we adopt a SE-attention module [47] to give the channels of feature attention weight, which enrich the feature maps and guide the model to focus on the salient fine-grained hand regions. The hierarchical strategy [48] reinforces the model at the multi-scale level, which make the model be robust for the different scale hands.

TABLE 3. Backbone of HA-SSD, 'DWConv' refers to the depth-wise convolution, 'S' refers to the stride of convolutional layer.

Type/Stride	Filter shape	Input size
Conv_1 / s2	$3 \times 3 \times 3 \times 32$	$300 \times 300 \times 3$
dwConv_2 / s1	$3 \times 3 \times 32$ dw	$150 \times 150 \times 32$
Conv_2 / s1	$1 \times 1 \times 32 \times 64$	$150 \times 150 \times 32$
dwConv_3 / s2	$3 \times 3 \times 64$ dw	$150 \times 150 \times 64$
Conv_3 / s1	$1 \times 1 \times 64 \times 128$	$75 \times 75 \times 64$
dwConv_4 / s1	$3 \times 3 \times 128$ dw	$75 \times 75 \times 128$
Conv_4 / s1	$1 \times 1 \times 128 \times 128$	$75 \times 75 \times 128$
dwConv_5 / s2	$3 \times 3 \times 128$ dw	$75 \times 75 \times 128$
Conv_5 / s1	$1 \times 1 \times 128 \times 256$	$38 \times 38 \times 128$
dwConv_6 / s1	$3 \times 3 \times 256$ dw	$38 \times 38 \times 256$
Conv_6 / s1	$1 \times 1 \times 256 \times 256$	$38 \times 38 \times 256$
dwConv_7 / s1	$3 \times 3 \times 256$ dw	$38 \times 38 \times 256$
Conv_7 / s1	$1 \times 1 \times 256 \times 512$	$38 \times 38 \times 256$
dwConv_8 / s1	$3 \times 3 \times 512$ dw	$38 \times 38 \times 512$
Conv_8 / s1	$1 \times 1 \times 512 \times 512$	$38 \times 38 \times 512$
Attention	-	$38 \times 38 \times 512$
dwConv_9 / s1	$3 \times 3 \times 512$ dw	$38 \times 38 \times 512$
Conv_9 / s1	$1 \times 1 \times 512 \times 512$	$38 \times 38 \times 512$
dwConv_10 / s2	$3 \times 3 \times 512$ dw	$38 \times 38 \times 512$
Conv_10 / s1	$1 \times 1 \times 512 \times 512$	$19 \times 19 \times 512$
dwConv_11 / s1	$3 \times 3 \times 512$ dw	$19 \times 19 \times 512$
Conv_11 / s1	$1 \times 1 \times 512 \times 512$	$19 \times 19 \times 512$
Attention	-	$19 \times 19 \times 512$
dwConv_12 / s2	$3 \times 3 \times 512$ dw	$19 \times 19 \times 512$
Conv_12 / s1	$1 \times 1 \times 512 \times 1024$	$10 \times 10 \times 512$
dwConv_13 / s1	$3 \times 3 \times 1024$ dw	$10 \times 10 \times 1024$
Conv_13 / s1	$1 \times 1 \times 1024 \times 1024$	$10 \times 10 \times 1024$
Attention	-	$10 \times 10 \times 1024$
Conv_14 / s1	$1 \times 1 \times 1024 \times 256$	$10 \times 10 \times 1024$
Conv_14 / s2	$3 \times 3 \times 256 \times 512$	$10 \times 10 \times 256$
Attention	-	$5 \times 5 \times 512$
Conv_15 / s1	$1 \times 1 \times 512 \times 128$	$5 \times 5 \times 512$
Conv_15 / s2	$3 \times 3 \times 128 \times 256$	$5 \times 5 \times 128$
Attention	-	$3 \times 3 \times 256$
Conv_16 / s1	$1 \times 1 \times 256 \times 128$	$3 \times 3 \times 256$
Conv_16 / s2	$3 \times 3 \times 128 \times 256$	$3 \times 3 \times 128$
Attention	-	$1 \times 1 \times 256$

After obtaining the 6 feature maps from the MobileNet structure, our goal is to increase the sensitivity of network to fine-grained informative features so that they can be exploited by detector. So, we propose to achieve this purpose by utilizing two operators named squeeze operator and excitation operator [47], which explicitly encode the channel inter-dependencies of feature maps in order to recalibrate filter responses before they are fed into prediction detector. A diagram of an HA building block is shown in Fig.4. The hierarchical channel-wise attention scheme is utilized to enrich these feature maps.

Due to standard convolutional layer neglecting to exploit the channel dependencies, we encode the global information

of each channel within the feature maps as the guidance of dependencies of the subsequent attention mechanism. Furthermore, the standard convolutional operator equips a local receptive field, which is unable to exploit contextual information outside of the region of kernel size. This issue is more severe in the light weight network due to the smaller receptive field sizes. To mitigate this problem, we adopt a squeeze operator which encode the global spatial information of the feature maps into a single channel representation. We achieve this purpose by utilizing global average pooling (GAP) [49].

Formally, given a feature map $u \in R^{H \times W \times C}$, we adopt a global average pooling layer along with the first and second dimensions to encode the global information as the input of the excitation operator:

$$z_c = F_{sq}(u_c) = \frac{1}{W \times H} \sum_{i=1}^W \sum_{j=1}^H u_c(i, j), \quad (3)$$

where $u = [u_1, u_2, \dots, u_c] \in R^{H \times W \times C}$ refers to the feature maps from the MobileNet and $z = [z_1, z_2, \dots, z_c] \in R^{1 \times 1 \times C}$ refers to the output of the global average pooling layer. F_{sq} represents to the GAP layer. We aim to fully capture channel-wise dependencies by aggregating the obtained global information into the squeeze operation. To fulfill this objective, it must learn the non-mutually exclusive relationship that allows multiple channels to emphasize against a hot activation. Hence, we use a simple-implement activation gating mechanism followed by sigmoid activated function $\sigma(\cdot)$:

$$s = F_{ex}(z, W) = \sigma(g(z, W)) = \sigma(W_2 \delta(W_1 z)), \quad (4)$$

where $s = [s_1, s_2, \dots, s_c] \in R^{1 \times 1 \times C}$ refers to the attention vector, δ represent to the ReLU function, W_1 and W_2 represent the parameters of the fully-connected layer. $W_1 \in R^{C/r \times C}$, where the r is a scaling parameter that is set to 16, $W_2 \in R^{C \times C}$. F_{ex} represents to the excitation operator.

The final output of this block is obtained by rescaling the transformation output with the activations:

$$\tilde{x}_c = F_{scale}(u_c, s_c) = s_c \cdot u_c, \quad (5)$$

where $\tilde{x}$ refers to the enriched feature map, F_{scale} represents to the element-wise multiply.

The hierarchical attention module enriches the outputs of DSConv8, DSConv11, DSConv13, Conv14, Conv15 and Conv16 layers to obtain the predicted bounding box maps B_i and the predicted classification confidence C_i at the multi-scale level. The rich features explore multi-scale contextual information and are able to focus on the salient regions. We adopt these predicted maps to calculate the loss of our network.

D. TRAINING OBJECTIVE

In this section, we introduce the loss functions used for training our proposed method. Formally, let c be the number of categories of hands, l represents the predicted bounding box and g refers to the ground truth bounding box. The overall

objective loss function is a weighted sum of localization loss (L_{loc}) and confidence loss (L_{conf}):

$$L(x, c, l, g) = \frac{1}{N} (L_{conf}(x, c) + \alpha L_{loc}(x, l, g)), \quad (6)$$

where α is the weight and set to 1, there are totally N matched boxes. If $N = 0$, the loss is set to 0. Specifically, the L_{loc} is Smooth L_1 loss followed by [19] between l and g .

We regress to the offsets of the center coordinate inside the bounding box and to predict its width and height:

$$L_{loc}(x, l, g) = \sum_{i=Pos}^N \sum_{\hat{g} \in \{c, cy, w, h\}} x_{ij}^k smooth_{L1}(l_i^m - \hat{g}_i^m), \quad (7)$$

$$\hat{g}_j^{cx} = (g_j^{cx} - d_i^{cx}) / d_i^w, \quad (8)$$

$$\hat{g}_j^{cy} = (g_j^{cy} - d_i^{cy}) / d_i^h, \quad (9)$$

$$\hat{g}_j^w = \log\left(\frac{g_j^w}{d_i^w}\right), \hat{g}_j^h = \log\left(\frac{g_j^h}{d_i^h}\right), \quad (10)$$

The confidence loss is the standard cross-entropy loss with softmax function :

$$L_{conf}(x, c) = - \sum_{i=Pos}^N x_{ij}^p \log(\text{softmax}(c_i^p)) - \sum_{i \in Neg} \log(\text{softmax}(c_i^p)), \quad (11)$$

V. EXPERIMENTS

In this section, we conduct experiments to evaluate the performance of hierarchical attention SSD on DSSL-Astronaut gesture dataset. To evaluate our approach more convincing, we also conduct some experiments on a public hand dataset named Oxford hand dataset.

Datasets: Our DSSL-Astronaut Gesture dataset aims at solving the gesture detection problem in the real space station. Our dataset contains totally 15126 images. The training set has 13840 images including all 6000 augmented images, while the rest 1286 real images serve as testing set. To verify the impact of artificial images while training the model, we also conduct the experiments on DSSL-Astronaut Gesture dataset without augmented images, which is named DSSLv0. The DSSLv0 dataset without augmented images consist of 7840 images for training and 1286 images for testing. As for the generality of model, we adopt 3024 real images and 6000 augmented images by one volunteer for training, and 6084 real images by other two volunteers for testing. This dataset is named single-DSSL. We also use only 3024 real images without augment images by the one volunteer for training, which is named single-DSSLv0. To evaluate the effectiveness of our proposed model more generic, we also adopt the public dataset named Oxford hand dataset where 4096 images for training, 738 images for validation and 821 images for testing. Specifically, hand instances in this dataset are annotated with one class label named 'hand'.

TABLE 4. Comparison result on Oxford hand dataset.

Method (Input Size)	mAP
SSD (300 × 300)	0.7491
YOLOv3 (320 × 320)	0.6952
HA-SSD (300 × 300)	0.7713

Implementation Details: Our method is based on pytorch framework with NVIDIA TITAN X. We adopt stochastic gradient descent (SGD) with an initial learning rate of 1e-03 and a momentum of 0.9 to train our model, the learning rate drops by 10 per 5000 steps. The input images of SSD and HA-SSD are resized to 300 × 300 × 3 resolution and the batches are set to 16. As follow the YOLOv3 [11], input images of YOLOv3 are resized to 320 × 320 × 3 and the batch size is set to 8.

Evaluation Metric: We adopt mean Average Precision (mAP) as our evaluation metric, which is widely used to evaluate the performance of the detector at an intersection-over-union (IOU) overlap with the threshold that is set to 0.5. mAP can be formulated as:

$$mAP = \frac{1}{Q} \sum_{q=1}^Q \frac{1}{n_q} \sum_{k=1}^{m_q} p_q(k) rel_q(k) \quad (12)$$

where Q is the number of gesture classes, n_q is the number of relevant documents, k is rank in the sequence of retrieved documents, m_q is all the number of all retrieved documents, $p_q(k)$ denotes the precision at rank k for the q -th query, $rel_q(k)$ denotes the relevance of prediction k for the q -th query, which is 1 if the k -th prediction is correct and 0 otherwise.

As to speed, frames per second (FPS) is used to measure detection speed for approaches.

A. PERFORMANCE COMPARISON ON OXFORD HAND DATASET

Table 4 shows the performance comparison of our HA-SSD against original SSD and YOLOv3 in terms of mAP on Oxford hand dataset. The input images of YOLOv3 are resized to 320 × 320 × 3 resolution and the input images of SSD are resized to 300 × 300 × 3 followed the original papers. We can see that our approach improves 2.2% mAP compared with original SSD and 7.6% compared with YOLOv3.

B. PERFORMANCE COMPARISON ON DSSL AND DSSLv0

We compare SSD, YOLOv3 and HA-SSD on the DSSL and DSSLv0 gesture dataset respectively. The testing results demonstrated in table 5 that all of three models achieve better performances on DSSL dataset than DSSLv0. This result indicates that augmented images as training data are beneficial for hand detection of the model. Our approach and YOLOv3 both outperform the SSD about 1.1% mAP improvement on DSSLv0. On DSSL gesture dataset, our HA-SSD also has slight improvement than SSD and is comparable with YOLOv3.

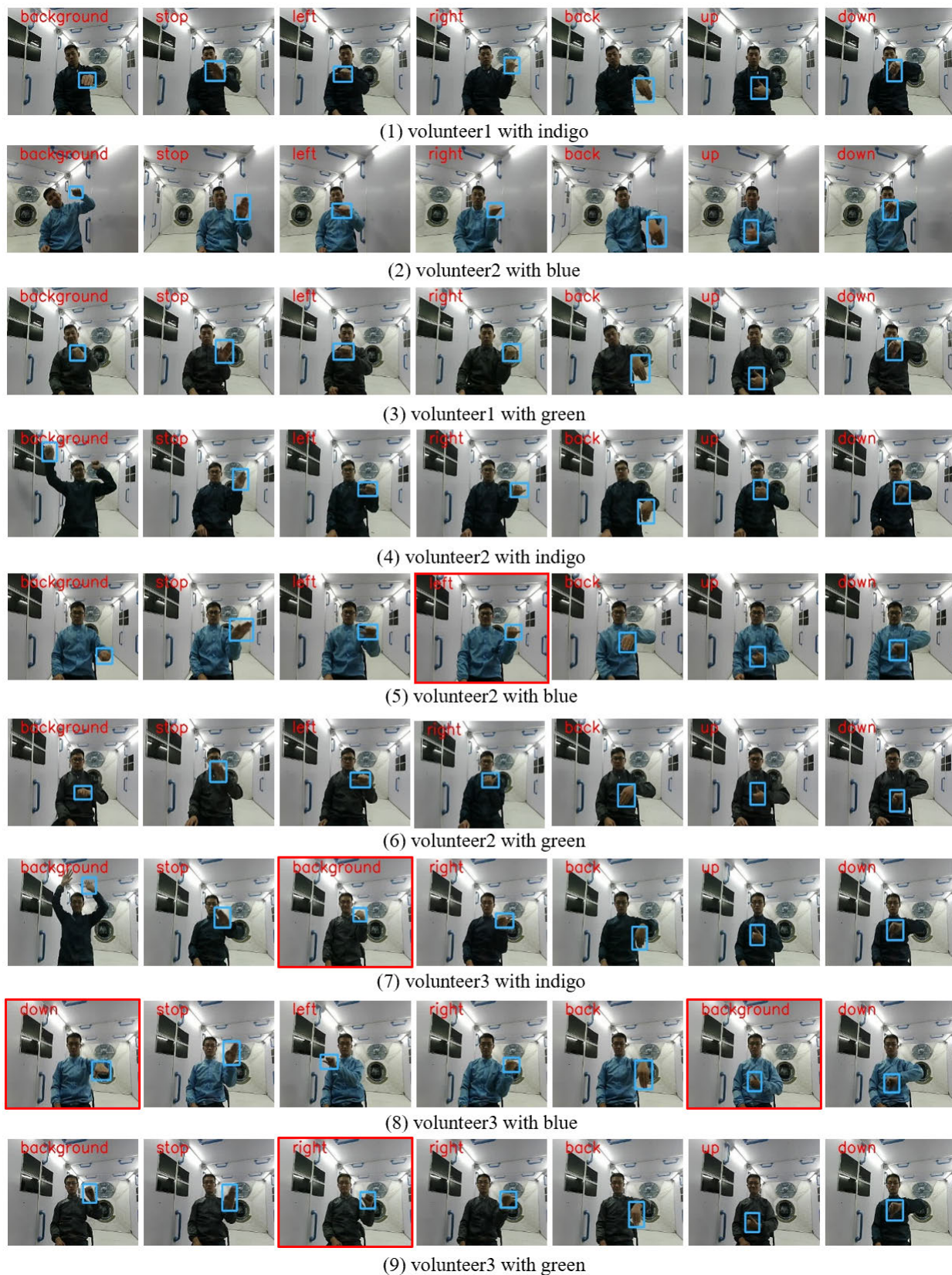


FIGURE 5. The visual recognition results. (1) (2) (3) are 7 gesture recognition results of volunteer 1 with indigo, blue and green, respectively; (4) (5) (6) are 7 gesture recognition results of volunteer 2 with indigo, blue and green, respectively; (7) (8) (9) are 7 gesture recognition results of volunteer 3 with indigo, blue and green, respectively. We mark the wrong results with a red border, such as the fourth image in (5), the third image in (7) and (9), the first and sixth image in (8).

TABLE 5. Comparison result on DSSLv0 and DSSL datasets.

Dataset	mAP		
	SSD	YOLOv3	HA-SSD
DSSLv0	0.9416	0.9512	0.9523
DSSL	0.9630	0.9730	0.9661

TABLE 6. Generalization test on single-DSSLv0 and single-DSSL datasets.

Dataset	mAP		
	SSD	YOLOv3	HA-SSD
Single-DSSLv0	0.7994	0.6255	0.8106
Single-DSSL	0.8853	0.5111	0.8968

TABLE 7. Speed comparison on DSSL-astronaut gesture dataset.

Method (Input Size)	Speed
SSD (300 × 300)	60FPS
YOLOv3 (320 × 320)	151FPS
HA-SSD (300 × 300)	142FPS

Generality is one of the most important factors in convolutional neural networks. It refers to that a well-trained model on a dataset is able to perform favourably on another similar dataset. In our mission, we hope that well-trained models on our dataset have good performance in the task of astronaut hand detection. Hence, we use SSD, YOLOv3 and HA-SSD to perform multiple experiments on single-DSSLv0 and single-DSSL datasets in the table 6. It shows that our approach is more generalized than SSD and YOLOv3 and adding augmented images into training data is able to improve the generalizability of SSD and HA-SSD.

A well-trained YOLOv3 has a good performance to detect hand on DSSL, but the performance of YOLOv3 on single-DSSL is poor. As authors of YOLOv3 refer in their paper, YOLOv3 struggles to get the boxes perfectly aligned with the object. In astronaut gesture recognition task, whether a finger is detected or not may lead to different recognition results, so the detection bounding boxes must perfectly enclose the entire hand of the astronaut. This result demonstrates YOLOv3 relies on training data too much and the model is easy to overfitting. YOLOv3 model performs worse on single-DSSL than single-DSSLv0. Because the augmented images are from the same volunteer's original images, and these augmented images exacerbate the overfitting of YOLOv3 model.

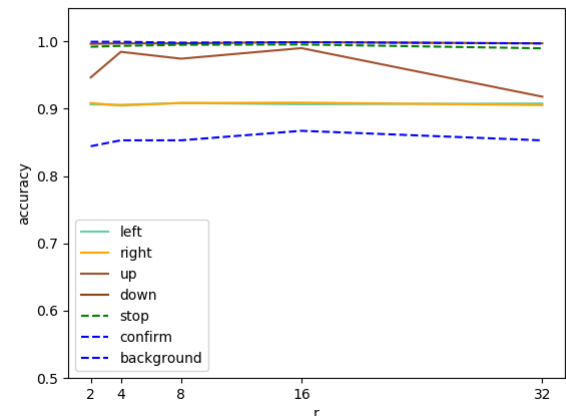
We measure the speed of model on the Titan X GPU, while the batch size is set to 16. The comparison of speed is shown in table 7 and we can find that HA-SSD model runs at 142 FPS under 300 × 300 resolution, which is 3× faster than original SSD. Even though this speed is slightly slower than fastest YOLOv3 in object detection, it is able to still be applied into the real scenario.

TABLE 8. Confusion matrix for HA-SSD on DSSL-astronaut gesture dataset, "BG" refer to the background.

Class	Left	Right	Up	Down	Stop	Confirm	BG
Left	0.9882	0.0059	0	0	0	0	0.0059
Right	0	1	0	0	0	0	0
Up	0	0	0.9907	0.0078	0	0	0.0015
Down	0	0	0	1	0	0	0
Stop	0	0	0	0	1	0	0
Confirm	0	0	0	0	0	1	0
BG	0.0034	0.0034	0.0101	0.0067	0.0034	0	0.9731

TABLE 9. Confusion matrix for HA-SSD on single-DSSL, "BG" refer to the background.

Class	Left	Right	Up	Down	Stop	Confirm	BG
Left	0.9375	0	0	0	0	0	0.0625
Right	0.2593	0.6667	0	0	0	0	0.0741
Up	0	0	0.8333	0	0	0	0.1667
Down	0	0.0769	0	0.8846	0	0	0.0385
Stop	0	0.0741	0	0	0.7037	0	0.2222
Confirm	0	0	0	0	0	0.9118	0.0882
BG	0.0161	0.0242	0.0161	0	0.0161	0.121	0.8065

**FIGURE 6.** Evaluation of The Proposed HA-SSD with Different Values of Parameter r .

We adopt confusion matrix to observe the recognition results of all gestures. Table 8 and Table 9 show that "background" gesture is more difficult to be recognized. In fact, we label some motion-blurred images as "background" if we deem they are too blurred to be recognized as meaningful gestures. Fig. 5 shows some notable visual recognition results, 3 volunteers with 3 colours clothes are shown. We mark the wrong results with red borders.

The comparison results show that our method achieves good performance which indicates that the HA-SSD and DSSL-Astronaut gesture dataset are very potential in future space station tasks. For HA-SSD, a possible improvement could be employed using a deburred and super-resolution model to process the images is able to boost the performance. For DSSL-Astronaut gesture dataset, we will collect more astronaut gestures data in the future with a variety of views, gestures, participants.

C. ABLATION STUDY

In equation (4), W_1 represent the parameters of fully-connected layer. $W_1 \in R^{C/r \times C}$, where scaling parameter r is a key parameter for the scaling level of global information, which controls the level to compress the global semantic information. We conduct experiments to evaluate the impact of r , the results are shown in Fig. 6. When the parameter r is set to 16, the performance of prediction is the best. It demonstrates that unsuitable parameters can lead to performance degradation, which is related to the distribution of the dataset.

VI. CONCLUSION

Real-time visual gesture recognition is an important component for the interaction between robot and astronaut in the space station. However, public gesture datasets are not suitable for astronaut gesture recognition and existing approaches trained on these datasets are not satisfied. In this paper, we introduce a new astronaut gesture dataset (DSSL-Astronaut gesture dataset) and a novel hierarchical attention single-shot detector network (HA-SSD) for astronaut gesture recognition. We conduct experiments to demonstrate that DSSL-Astronaut gesture dataset is suitable for astronaut gesture recognition. HA-SSD is able to effectively localize and recognize the gestures of astronauts with strong generality. More astronaut gestures data will be collected in the future with a variety of views, gestures, participants and data formats (e.g. depth images). More data augmentation techniques (e.g. DCGAN [50], etc.) will be considered. The experiments will be conducted by Nvidia Jetson TX2 in the future.

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